VANSH AGRAWAL

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B.Tech, Computer Science and Engineering — AI & ML
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PROFILE

AI/ML Engineer with hands-on experience building and deploying production-grade machine learning systems, specializing in computer vision and real-time applications. Currently working as an AI/ML Intern at Tata Steel, developing scalable solutions for industrial use cases. Built multiple end-to-end projects across healthcare AI, predictive systems, and real-time communication, with a strong focus on practical impact and deployable architectures.

SKILLS

Languages
Python · JavaScript · SQL · Java · C / C++ · HTML5 / CSS3

Frameworks & Libraries
TensorFlow · OpenCV · XGBoost · FastAPI · Flask · React 19 · Node.js · WebRTC · Scikit-learn · Pandas · NumPy

Tools & Databases
Git / GitHub · Docker · Firebase · MySQL · Vercel · REST APIs · Postman · CI / CD

EXPERIENCE

Tata Steel
Nov 2025 – Present

AI / ML Intern — *Chennai, India*

— Engineered a Computer Vision pipeline using OpenCV for automated TMT rebar quality control, calculating rib parameters with 95% precision to evaluate strict pass/fail tolerances.
— Developed an automated validation module for chemical ring tests, utilizing spatial analysis to accurately measure rod thickness and reduce manual inspection time by 40%.

RCOTBA Industries
Jan 2025 – Present

Technical Manager (Promoted from Web Developer) — *Chennai, India*

— Promoted through 3 leadership tiers (Web Developer → Tech Manager) by directing cross-functional teams and hitting 100% of agile delivery deadlines.
— Developed custom client websites using React, streamlining internal pipelines to deploy new sites from concept to live hosting in just 2-3 days and increasing delivery speed by 35%.

PROJECTS

AERIS — Flood Risk Intelligence System
GitHub

XGBoost · Python · FastAPI · React 19 · GeoJSON

— Developed AERIS, an AI-powered predictive flood management platform for the Bihar region, utilizing a React/Tailwind frontend and a FastAPI backend to deliver proactive 7-day disaster response intelligence. Integrated dual XGBoost models and a React-Leaflet GIS dashboard to analyze live telemetry across 18 river stations, simulating multi-variate hydraulic stress and automating breach predictions.

LitterVision — Smart City Waste AI
GitHub

TensorFlow · OpenCV · Python · Flask

— Developed LitterVision, an image classification system using MobileNetV2 and Transfer Learning, achieving 78% accuracy across 6 categories of urban waste. Implemented a confidence-based filtering system (0.60 threshold) and a two-stage fine-tuning pipeline to ensure reliable, high-precision predictions in real-world deployment.

Voxera — Low-Latency P2P Video
GitHub

WebRTC · Node.js · Firebase · JavaScript

— Developed Voxera, a high-performance peer-to-peer communication engine enabling multi-user voice and 4K screen sharing. Engineered a WebRTC Peer-Mesh network seamlessly integrated with Firebase signaling to minimize overhead, achieving <150ms end-to-end media latency globally.

EDUCATION

SRM IST, Kattankulathur
B.Tech — CSE (AI & ML)
2023 – 2027 · CGPA: 7.7 / 10

A.P.J School, NOIDA
Class XII — CBSE (MPC)
2023 · 68.4%

A.P.J School, NOIDA
Class X — CBSE
2021 · 85.2%